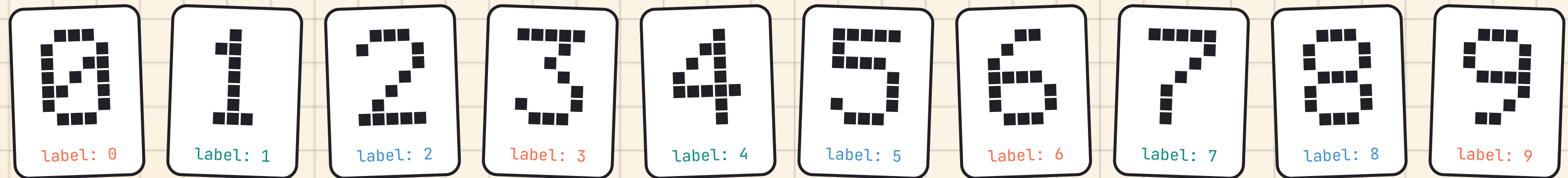


Our Exercise: Recognize Handwritten Digits

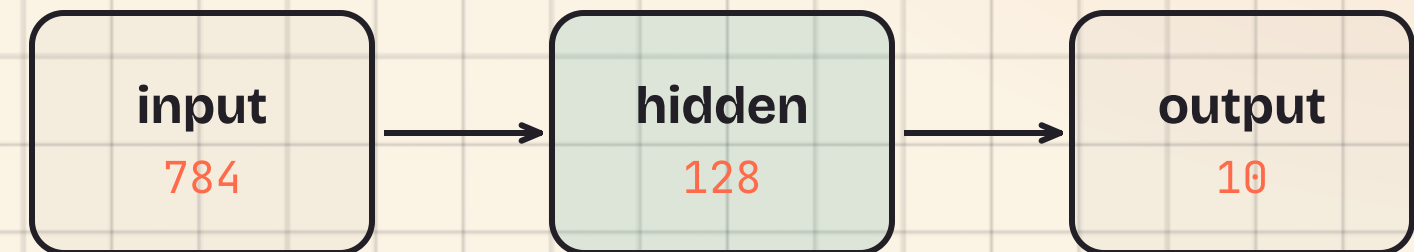
MINIST is 70,000 handwritten digits, each a 28×28 grayscale image labeled 0 through 9. It's the classic first dataset for computer vision.



Open notebook in Colab ↗

Let's Build It: **A Tiny Net** From Scratch

Two layers of weights, two activation steps — that's the whole brain.



simple_net.py

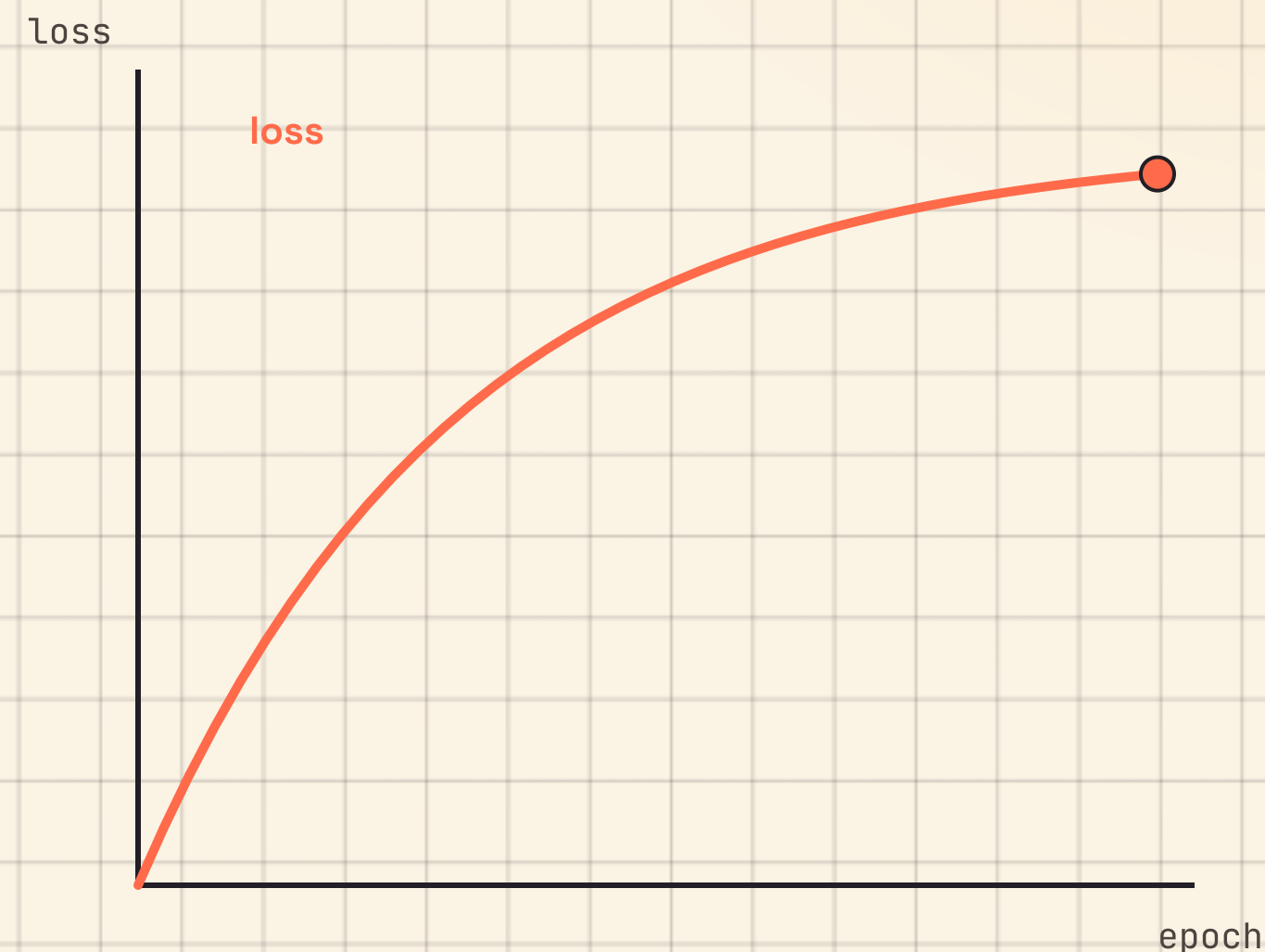
```
class SimpleNet:
    def __init__(self):
        self.W1 = random_matrix(784, 128)
        self.W2 = random_matrix(128, 10)
    def forward(self, x):
        h = relu(x @ self.W1)
        return softmax(h @ self.W2)
```

Train By Repeating: Predict, Measure, Adjust

Thousands of tiny corrections later, the loss curve drops and the network gets sharper.

train.py

```
for epoch in range(epochs):  
    predictions = net.forward(X_train)  
    grads = backprop(predictions, y_train)  
    net.update(grads, learning_rate)
```



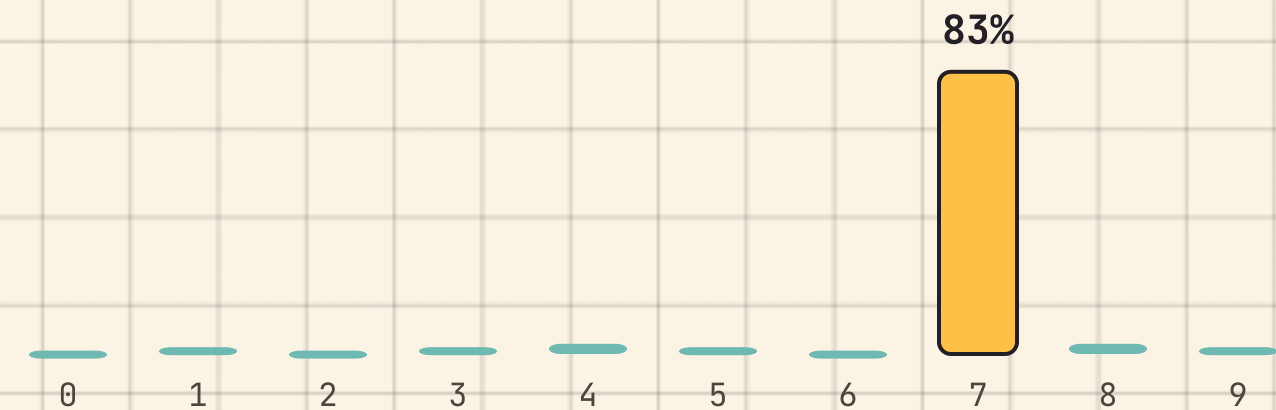
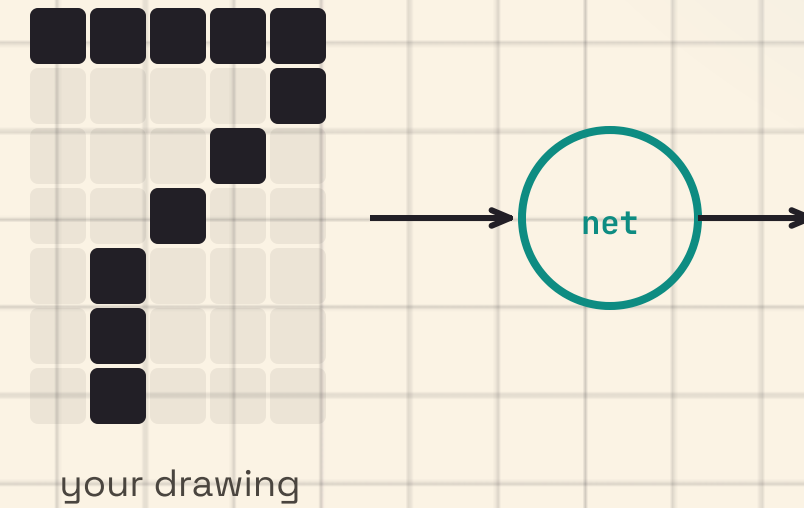
← back to the deck

Show It a New Digit — What Does It See?

The trained network turns a fresh drawing into ten confidence scores, then picks the highest one.

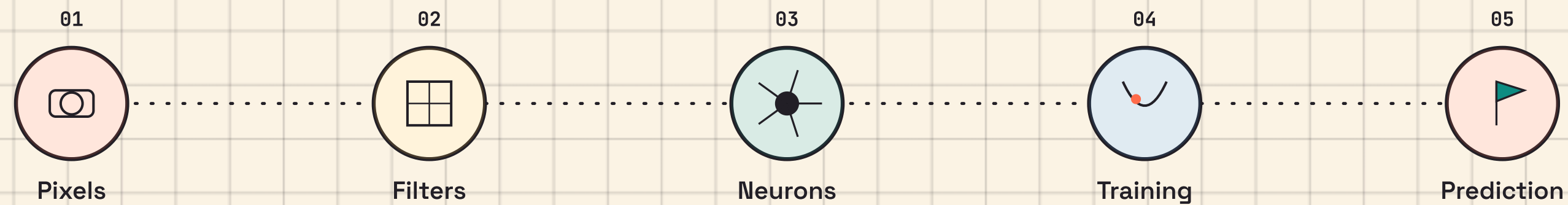
 predict.py

```
prediction = net.forward(my_drawing)
digit = argmax(prediction)  # 7
```



Nice work today

You Just Built the Core of **Computer Vision**



Pixels are numbers

Filters find features

Neurons combine evidence

Training shrinks error